MA6212 Group Project Report

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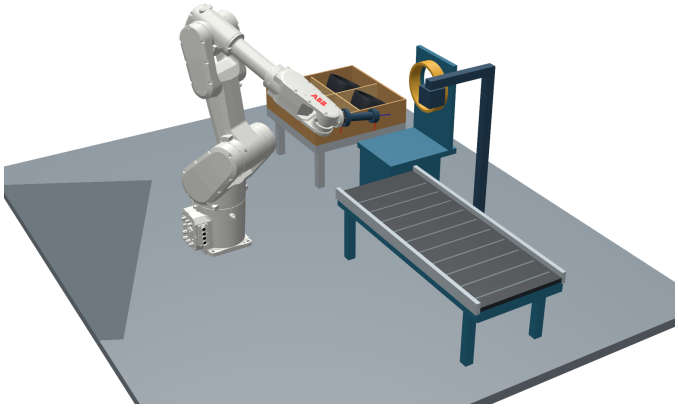


Fig. 1. Completed workcell as rendered in RobotStudio.

The project code, raw data, simulation files, and videos (excluding the group photographs) are available at https://github.com/MrOxOrOxen/ntu/tree/ma6212.

The group photographs are included both in the ZIP file submitted through NTULearn and at the end of this report.

I. Introduction

This project reproduces the elbow-pipe handling sequence shown in the reference video. The robot picks up a pipe from the conveyor and inserts it into a horizontal port at the gasket station. After releasing the pipe, it grasps it again at a different contact point and transfers it to the assigned position in a packing box.

Three challenges shaped our approach: the pipe model needed to accommodate two different grasp poses, the pickup target had to adapt to the incoming pipe pose, and the control program had to coordinate robot motion with each attachment and release.

We selected the ABB IRB 1300-7/1.4 from the official robot library and built the surrounding equipment using the RobotStudio SDK. The cell runs on an OmniCore virtual controller with RobotWare 7.21.5. Although the assignment brief specifies IRC5, the installed IRB 1300 template is configured for OmniCore, so we followed the controller configuration supported by that model.

The system is organized into three parts. C# scripts generate the cell geometry and nominal robot targets, RAPID handles motion and process sequencing, and a Smart Component updates the workpieces and logs the simulation results. The completed cell is shown in Fig. 1.

TABLE I
Main Equipment and Process Parameters

Item	Value or implementation
Robot	IRB 1300-7/1.4; 7 kg payload; 1.4 m reach
Software	RobotStudio 2026.3; RobotWare 7.21.5
Pipe	OD 215 mm; wall 15 mm; height 150 mm; mass 2.5 kg
Tool	TCP offset 180 mm; assumed mass 1.0 kg
Packing	Four positions per box; 270 mm pitch
Batch	Four parts; one box
Vision range	$\Delta x, \Delta y \in [-15, 15]$ mm; $\psi \in [-5, 5]^\circ$
Process waits	Belt feed: 3 s/part; gasket: 2 s/part

TABLE II
Elbow Geometry Parameters

Parameter	Symbol	Value
Outer radius	r_o	107.5 mm
Inner radius	r_i	92.5 mm
Wall thickness	t	15 mm
Overall height	H	150 mm
Selected bend angle	θ	30°
Centerline bend radius	R	192.5 mm
Calculated solid volume	V	949,949 mm^3

II. Geometric Modeling

A. Parametric Elbow Model

The assignment specifies the pipe's outside diameter, wall thickness, and overall height, but not its bend angle. We therefore selected a bend angle of 30° based on the reference video. The resulting geometric parameters are listed in Table II.

We began by creating two circular surfaces with radii r_o and r_i in the local XY plane. Subtracting the smaller surface from the larger one produced an annular cross-section, which we then swept around an axis passing through $(R, 0, 0)$ and parallel to the local Y axis. The resulting hollow elbow is shown in Fig. 2.

The outer bend radius determines the overall height:

$$H = (R + r_o) \sin\theta. \tag{1}$$

Using the specified height, we can then solve for R:

$$R = \frac{H}{\sin\theta} - r_o = 192.5 \text{ mm}. \tag{2}$$



Fig. 2. Hollow elbow obtained by sweeping an annular cross-section through the selected 30° bend.

TABLE III
TOOL GEOMETRY ALONG ITS LOCAL Z AXIS

Section	Radius (mm)	Length (mm)	Z range (mm)
Flange connector	40	18	0–18
Shaft	18	130	18–148
Head	32	25	148–173
Contact end	25	7	173–180

The complete swept geometry can be described in terms of the section radius ρ, section angle ϕ, and sweep angle α:

$$x = R + (\rho\cos\phi - R)\cos\alpha, \tag{3}$$

$$y = \rho\sin\phi, \tag{4}$$

$$z = (R - \rho\cos\phi)\sin\alpha, \tag{5}$$

Here, $r_i \le \rho \le r_o$, $0 \le \phi < 2\pi$, and $0 \le \alpha \le \theta$. At $\alpha = 0$, the cross-section lies in the XY plane; increasing α moves it along the bend.

The model was generated using *CreateSurfaceCircle*, *Cut2*, and *Body.Sweep*. Its theoretical volume is the annular cross-sectional area multiplied by the centerline arc length:

$$V = \pi(r_o^2 - r_i^2)R\theta. \tag{6}$$

This gives a volume of approximately 0.950 L. The required mass was assigned separately in RAPID.

B. Vacuum Tool and TCP

The vacuum tool consists of four coaxial cylindrical sections, with dimensions given in Table III. Its local origin is at the mounting flange, and the contact end extends along the positive Z axis.

With this geometry, the TCP is located at $[0, 0, 180]^T$ mm in the flange frame, with no additional rotation between the tool and flange frames. We used *AttachToFlange* to align the tool origin with the flange in the final assembly, and applied the same transform in the RAPID tool data.

TABLE IV
DIMENSIONS OF THE CREATED CELL MODELS

Component	Dimensions and placement (mm)
Conveyor body	$440 \times 1100 \times 50$
Conveyor surface	$420 \times 1100 \times 20$; top at $z = 350$
Conveyor legs	Four $40 \times 40 \times 280$ blocks
Fixture base	$300 \times 300 \times 320$
Fixture top	$400 \times 320 \times 30$; top at $z = 350$
Wall panel	$25 \times 320 \times 460$; port centre at $z = 650$
Port collar	OD 260; ID 232; axial length 60
Box bottom	$550 \times 550 \times 20$; top at $z = 300$
Box walls	Thickness 10; top at $z = 420$
Box dividers	Two crossed boards; thickness 8; height 110
Box support	$570 \times 570 \times 50$ top; four legs
Camera structure	$40 \times 40 \times 1000$ post; $340 \times 40 \times 40$ arm
Camera housing	$80 \times 80 \times 80$
Reference gasket	OD 200; ID 185; thickness 5

We assigned the tool a mass of 1 kg and assumed its centre of gravity to be at $[0, 0, 80]^T$ mm in the tool load frame. The pipe load is 2.5 kg. In RAPID, *GripLoad loadPipe* is applied after grasping the pipe and *GripLoad load0* after releasing it.

The combined carried mass is therefore 3.5 kg. The load centres and moments of inertia are estimates used for this handling simulation.

C. Cell Equipment

We retained the geometry and kinematic model supplied with the ABB robot. The conveyor, fixture, packing box, support, and camera were constructed from simple solid shapes. Their final dimensions are summarized in Table IV.

The fixture consists of a vertical panel with a horizontal port. Its collar faces the robot in the negative world X direction. During insertion, the pipe origin moves from $x = 1020$ to 1100 mm. As the collar begins at $x = 1080$ mm, the final insertion depth is 20 mm. The robot then releases the pipe and allows 2 s for the gasket process. Following the second grasp, it withdraws the pipe by 80 mm. Both strokes use *v50, fine*.

Each pipe travels on the conveyor from $y = -1250$ mm to $y = -550 + \Delta y$ mm over 3 s, after which the belt stops for pickup. The measured x position and yaw remain unchanged during feeding. The y position is given by

$$y(t) = -1250 + (700 + \Delta y)\min(t/3, 1), \quad t \ge 0. \tag{7}$$

Here, $t = 0$ marks the start of belt feeding.

III. FRAMES AND GRASP POSE CALCULATION

A. Frame Definitions

We define four coordinate frames: the world frame W, pipe frame O, robot flange frame F, and TCP frame C. The world origin coincides with the robot base, while the pipe origin is at the centre of its initial annular cross-section. The RAPID workobject *wCell* coincides with the world frame.

TABLE V
LOCAL CONTACT POSES

Quantity	Contact A	Contact B
x (mm)	-102.942326	-79.392336
y (mm)	0	0
z (mm)	52.094453	126.785479
Tool pitch	100°	115°
Quaternion scalar	0.642788	0.537300
Quaternion y	0.766044	0.843391

Using ${}^{A}T_{B}$ to denote the transformation from frame B to frame A, we calculate the required contact pose as

$$ {}^{W}T_{C} = {}^{W}T_{O}\,{}^{O}T_{C}. \tag{8}$$

The flange pose is then obtained from

$$ {}^{W}T_{F} = {}^{W}T_{C}({}^{F}T_{C})^{-1}, \tag{9}$$

where ${}^{F}T_{C}$ represents a 180 mm translation along the flange Z axis.

B. Two Surface Contacts

Both grasp points were selected on the outer curved surface of the elbow. A point on this surface can be expressed as

$$ {}^{O}\mathbf{p}_{C}(\alpha) = \begin{bmatrix} R - (R + r_o)\cos\alpha \\ 0 \\ (R + r_o)\sin\alpha \end{bmatrix}. \tag{10}$$

We chose $\alpha_A = 10^\circ$ for contact A and $\alpha_B = 25^\circ$ for contact B. Their local poses are given in Table V.

For either contact, the tool orientation is defined by

$$ {}^{O}R_{C} = R_y(90^\circ + \alpha). \tag{11}$$

The resulting positive tool Z direction is $[\cos\alpha, 0, -\sin\alpha]^T$, pointing inward along the local surface normal. The quaternion components not shown in Table V are zero.

The separation between the two contact points is

$$ d_{AB} = \|{}^{O}\mathbf{p}_{B} - {}^{O}\mathbf{p}_{A}\|_2 \tag{12}$$

$$ = 2(R + r_o)\sin\left(\frac{\alpha_B - \alpha_A}{2}\right) \approx 78.32 \text{ mm}. \tag{13}$$

The corresponding surface normals are separated by 15°.

C. Simulated Vision Input

At the pickup point, the nominal pipe origin is $[650, -550, 350]^T$ mm. We represent variations in the incoming pose using planar offsets Δx and Δy, together with a yaw angle ψ:

$$ {}^{W}T_{O} = \begin{bmatrix} R_z(\psi) & [650 + \Delta x, -550 + \Delta y, 350]^T \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{14}$$

To make the simulated vision inputs reproducible, we use the deterministic generator

$$ s_{k+1} = (17 s_k + 43) \bmod 997, \qquad s_0 = 317, \tag{15}$$

$$ u_k = s_k / 996. \tag{16}$$

TABLE VI
RECONSTRUCTED VISION INPUTS FROM THE PROGRAM SEED

Part	Δx (mm)	Δy (mm)	ψ (deg)
1	-1.5361	4.9699	-1.5161
2	13.8253	-4.1566	-3.1827
3	-11.1145	-7.7108	-3.3032
4	12.7711	7.9518	-4.6386

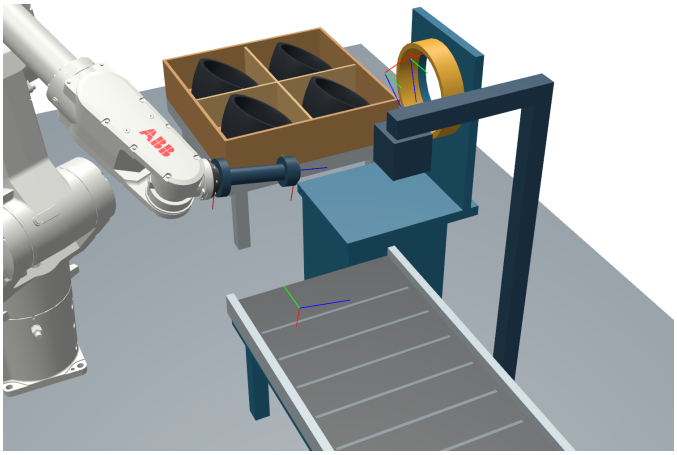


Fig. 3. Nominal pickup frame (left) and fixture contact frames A and B (right). The camera is represented geometrically as an ideal pose source.

Three successive samples define each incoming pose:

$$ \Delta x = 30u_1 - 15, \qquad \Delta y = 30u_2 - 15, \tag{17}$$

$$ \psi = 10u_3 - 5. \tag{18}$$

The translation offsets are measured in millimetres, and yaw is measured in degrees.

Applying (8) to contact A gives the following TCP position:

$$ x_C = 650 + \Delta x - 102.942326\cos\psi, \tag{19}$$

$$ y_C = -550 + \Delta y - 102.942326\sin\psi, \tag{20}$$

$$ z_C = 402.094453. \tag{21}$$

The corresponding orientation is $R_z(\psi)R_y(100^\circ)$. RAPID applies this orientation through *OrientZYX(visionYaw,100,0)*, while the Smart Component uses the same offsets to place the CAD pipe.

The resulting input sequence is shown in Table VI. For the first part, the pipe origin is approximately $(648.464, -545.030, 350)$ mm. Fig. 3 illustrates the nominal contact frames.

IV. TARGET GENERATION AND MOTION PLANNING

A. Layout and Nominal Targets

The overall layout is shown from above in Fig. 4, with the corresponding object origins listed in Table VII.

Combining these object origins with the local contact poses gives the nominal TCP targets in Table VIII. For pickup

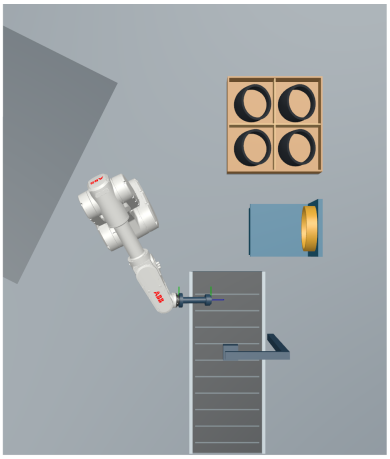


Fig. 4. Top view of the workcell. All coordinates are expressed in the robot base frame.

TABLE VII
OBJECT ORIGINS IN THE WORLD FRAME

Location	x (mm)	y (mm)	z (mm)
Nominal pickup	650	−550	350
Intermediate fixture	1100	−150	650
Box slot 1	1070	350	300
Box slot 2	1070	620	300
Box slot 3	800	350	300
Box slot 4	800	620	300

and box placement, the approach and clearance positions are defined as

$$\mathbf{p}_{approach} = \mathbf{p}_{contact} + [0, 0, 160]^T, \tag{22}$$

$$\mathbf{p}_{clear} = \mathbf{p}_{contact} + [0, 0, 380]^T. \tag{23}$$

Both offsets are applied along world Z, rather than tool Z. At the wall port, we use the pipe rotation $R_y(-90^\circ)R_z(180^\circ)$ so that the inlet faces the wall while the bend points upward. The fixture approach and clearance offsets are $[-200, 0, 0]^T$ and $[-300, 0, 300]^T$ mm, respectively, in world coordinates. In total, seven contact groups define 21 nominal targets. Two further offsets of $[-80, 0, 0]^T$ mm define the slow insertion and withdrawal strokes.

B. Inverse Kinematics and Configuration Choice

For the fixture contacts, the TCP quaternions are $[0, 0.996195, 0, 0.087156]$ at A and $[0, 0.976296, 0, 0.216440]$ at B.

For each pose, the C# script creates an *RsRobTarget* and an *RsTarget*, then calls *GetAllConfigurationsAsync* using the tool definition. We retain the inverse-kinematics solutions with *cfx=1* and select the one that minimizes

$$\mathbf{q}^* = \arg\min_{\mathbf{q}\in\mathcal{C}_1} \sum_{j=1}^{6} q_j^2, \tag{24}$$

TABLE VIII
NOMINAL TCP CONTACT TARGETS

Target	x (mm)	y (mm)	z (mm)	Orientation
Pickup A	547.058	−550	402.094	100°
Fixture A	1047.906	−150	752.942	Wall pose A
Fixture B	973.215	−150	729.392	Wall pose B
Box 1	990.608	350	426.785	115°
Box 2	990.608	620	426.785	115°
Box 3	720.608	350	426.785	115°
Box 4	720.608	620	426.785	115°

TABLE IX
JOINT SOLUTION AT THE NOMINAL PICKUP CONTACT

q_1	q_2	q_3	q_4	q_5	q_6
−62.924	22.107	49.467	−67.101	−72.156	17.198

All angles are in degrees.

where $\mathcal{C}_1$ denotes the retained configuration set. This procedure provides a reference joint configuration for each nominal target.

We found a valid solution in this set for every final nominal target. Table IX gives one solution for the nominal pickup contact. Each RAPID target stores the position, quaternion, configuration, and external-axis fields.

C. Motion Between Targets

During motion, the TCP pose is related to the flange pose by

$$^{W}T_C(t) = {}^{W}T_F(\mathbf{q}(t))\,{}^{F}T_C. \tag{25}$$

RobotWare handles joint motion and interpolation, so the motion-planning task here is to specify suitable targets, motion instructions, speeds, and zones.

Near the workpiece, we use *MoveL* to follow a linear TCP path. The position along such a segment can be written as

$$\mathbf{p}(s) = \mathbf{p}_0 + s(\mathbf{p}_1 - \mathbf{p}_0), \quad 0 \le s \le 1. \tag{26}$$

The tool orientation remains fixed along the approach and lift segments, while *MoveJ* is used to move between elevated targets.

We use *fine* at contact targets to require a stop at the specified pose. Selected intermediate targets use *z10* or *z20* to allow smoother transitions without stopping at each point. The motion settings are summarized in Table X. For the tested sequence, *ConfJ* is enabled and *ConfL* is disabled.

V. RAPID AND SMART COMPONENT IMPLEMENTATION

A. Motion Sequence

The main routine begins by moving the robot to the pickup clearance position, after which it starts the batch clock. The operations within each cycle are listed in Table XI, and the overall batch logic is shown in Fig. 5. The following RAPID excerpt covers the initial pickup and placement at the fixture.

TABLE X
PROGRAMMED MOTION SETTINGS

Motion or setting	RAPID value
Main transfer	*v500*
Approach	*v200* or *v300*
Contact	*v100, fine*
Insertion / withdrawal	*v50, fine*
Lift	*v200*
Intermediate zones	*z10*, *z20*
Speed setting	*VelSet 100,800*
Acceleration setting	*AccSet 60,60*

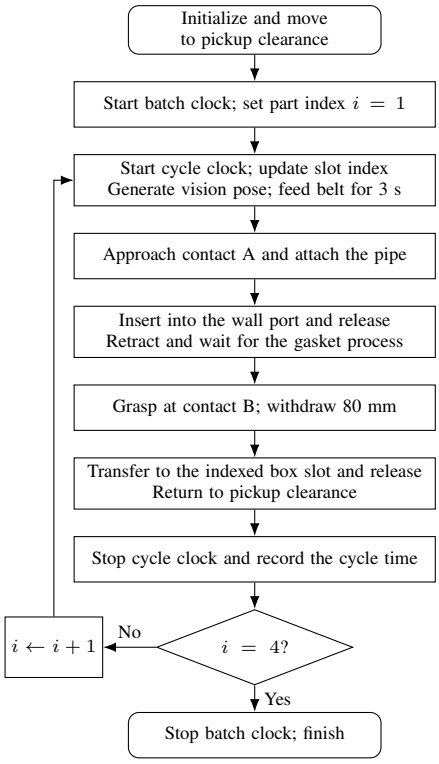


Fig. 5. Control flow for the four-part packing batch.

```
MoveJ Offs(pSeen,0,0,380),v500,z20,tVac\WObj:=wCell;
MoveL Offs(pSeen,0,0,160),v300,z10,tVac\WObj:=wCell;
MoveL pSeen,v100,fine,tVac\WObj:=wCell;
Stage 20;
GripLoad loadPipe;
MoveL Offs(pSeen,0,0,380),v200,z10,tVac\WObj:=wCell;
MoveJ pFixtureAClear,v500,z20,tVac\WObj:=wCell;
MoveL pFixtureAApproach,v300,z10,tVac\WObj:=wCell;
MoveL Offs(pFixtureAContact,-80,0,0),v100,fine,tVac\WObj:=
    wCell;
Stage 25;
MoveL pFixtureAContact,v50,fine,tVac\WObj:=wCell;
Stage 30;
GripLoad load0;
```

Once the pipe has been inserted and released at the fixture, the tool retracts and waits for the gasket process to finish. It then approaches contact B for the second grasp. The complete RAPID module is included in the virtual-controller backup within the Pack & Go file.

TABLE XI
ONE-PART WORKFLOW

Step	Operation
1	Start cycle clock; enter the inter-part state.
2	Update indices and vision pose; feed the belt for 3 s.
3	Approach contact A and attach the pipe.
4	Lift, transfer, and insert the pipe into the wall port.
5	Release, retract, and wait 2 s for the gasket process.
6	Grasp at contact B and withdraw the pipe by 80 mm.
7	Transfer to the selected box slot and release.
8	Retract and return to pickup clearance.
9	Stop cycle clock; finish after four parts.

TABLE XII
VARIABLES SHARED WITH THE SMART COMPONENT

Variable	Purpose
nStage	Process state
nPart, *nSlot*	Current part and box slot
visionX, *visionY*	Planar input offsets
visionYaw	Input rotation about world Z
batchComplete	Batch completion flag
cycleTimes{4}	Per-part controller times
totalTime	Total controller time

B. Box Indexing

For each part i, the corresponding box slot is determined by

$$k = ((i-1) \bmod 4) + 1. \tag{27}$$

The origins of the four box slots can be calculated using

$$c = \left\lfloor \frac{k-1}{2} \right\rfloor, \qquad r = (k-1) \bmod 2, \tag{28}$$

$$x_{box} = 1070 - 270c, \qquad y_{box} = 350 + 270r, \tag{29}$$

$$z_{box} = 300. \tag{30}$$

With this indexing order, the far column is filled before the near column. RAPID uses *TEST nSlot* to choose among the four contact targets, and the batch finishes once all four positions are occupied.

C. State Exchange

The Smart Component inherits from *SmartComponentCodeBehind* and communicates with the local virtual controller through the PC SDK. When the simulation starts, it retrieves references to the tool, the four pipe instances, and the RAPID variables listed in Table XII.

At every simulation step, the component reads the current process state and part index. Changes in these values trigger the actions listed in Table XIII. The RAPID *Stage* procedure updates *nStage* and waits for 0.10 s, giving the component time to detect the new state.

We introduced state 5 after examining the event log. Without this intermediate state, the part index could be updated while the process was still in release state 50, causing the Smart Component to attribute the previous release to the next part.

TABLE XIII
PROCESS STATES AND GEOMETRY ACTIONS

State	Action
0	Hide pipe instances and remove tool attachments.
5	Leave the previous release state before changing part identity.
8	Move the pipe along the belt for 3 s.
10	Stop at the simulated vision pose.
20	Attach the pipe at contact A.
25	Start the 80 mm insertion move.
30	Release at the fixture and record position error.
40	Attach the pipe at contact B.
45	Complete the 80 mm withdrawal.
50	Release in the box and record position error.

In the final program, state 5 is entered before the indices are updated. This ensures that each part has one recorded fixture release and one recorded box release.

D. Attachment and Release

Let G be the tool component frame. At the instant of grasping, t_g, the pipe pose relative to this frame is

$$ {}^{G}T_{O} = ({}^{W}T_{G}(t_g))^{-1}\, {}^{W}T_{O}(t_g). \tag{31} $$

Once the pipe is attached, this relative transform remains fixed. During transport, its world pose is therefore

$$ {}^{W}T_{O}(t) = {}^{W}T_{G}(t)\, {}^{G}T_{O}. \tag{32} $$

The attachment is implemented with the following SDK call:

```
tool.Attach(current, false,
            Matrix4.Identity);
```

This preserves the pipe's world pose at the moment of attachment, after which it moves with the tool.

To release the pipe, the code saves its *GlobalMatrix*, detaches it from the tool, and restores the saved matrix so that its world pose is preserved.

A single box is represented by four pipe instances. The Smart Component initially hides all four and makes each instance visible when its conveyor feed begins.

VI. MEASUREMENT METHODS

A. Controller Timing

Timing is recorded using *batchClock* for the full batch and *cycleClock* for individual parts. The batch clock starts after the robot reaches the initial pickup clearance position and stops after its final return. Each cycle time includes conveyor feeding, the gasket-process dwell, and the state-handshake waits.

The mean cycle time and throughput are calculated as

$$ \overline{T} = \frac{1}{N}\sum_{i=1}^{N} T_i, \tag{33} $$

$$ Q = \frac{3600N}{T_{batch}}. \tag{34} $$

Both metrics are based on the virtual controller's clock readings.

TABLE XIV
MEASURED RESULTS FOR THE FINAL BATCH

Part	Slot	Cycle (s)	Fixture error (mm)	Box error (mm)
1	1	37.228	0.000123	0.000266
2	2	38.252	0.000119	0.039227
3	3	37.088	0.577977	0.586858
4	4	38.028	0.156165	0.191246

B. Position Error

We evaluate placement error by comparing the pipe origin with its target position after release at the fixture and in the box:

$$ e_i = \|\mathbf{p}_{actual,i} - \mathbf{p}_{target,i}\|_2. \tag{35} $$

The Smart Component converts the SDK coordinates from metres to millimetres and records the errors at states 30 and 50. We also report the root-mean-square (RMS) error:

$$ e_{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2}. \tag{36} $$

These measurements cover positional accuracy only; a separate orientation-error metric was not recorded.

C. Collision and TCP Recording

The simulation uses a requested step size of 24 ms. At each step, *CollisionDetector.CheckCollision* checks the robot, tool, and carried pipe against the conveyor, fixture, port collar, packing box, dividers, camera structure, and box support. The carried pipe is also checked against any visible pipes already in the box.

The collision checks exclude intentional contact between the tool and pipe. Contact with the fixture support is excluded when the pipe is within 1 mm of its expected object origin. Conveyor support contact is excluded below $z = 351$ mm within the feed region $-1260 \le y < -520$ mm. Checks involving the port collar and box dividers remain active during insertion and placement. Robot self-collision and contact with the floor are not covered by this custom check.

The TCP position is recorded using

$$ \mathbf{p}_{TCP}(t) = \mathrm{trans}\left[{}^{W}T_{G}(t)\,\mathrm{Trans}(0,0,180)\right], \tag{37} $$

with consistent units for translation. The final four-part batch produced 6,286 TCP samples. To make Fig. 6 easier to read, we plotted every third sample and omitted repeated points during stationary periods. The complete trace is also retained as a hidden component in the station.

VII. RESULTS AND DISCUSSION

A. Per-Part Results

The measured cycle times and release errors for each part are reported in Table XIV. All four pipes completed the insertion, withdrawal, fixture-release, and box-release stages.

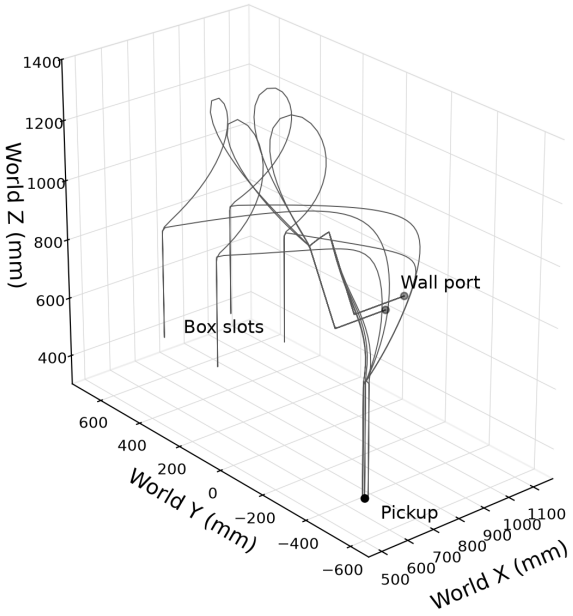


Fig. 6. Measured TCP trace from the full batch.

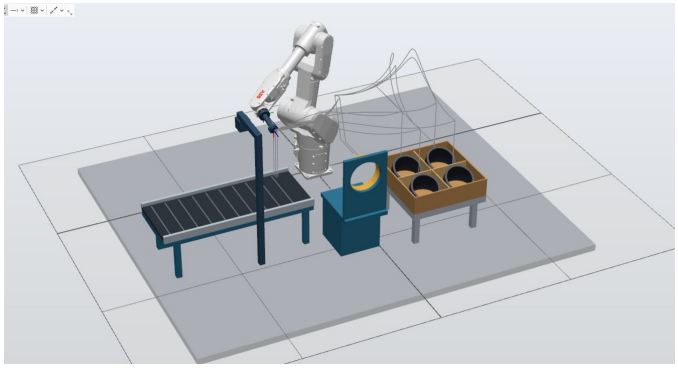


Fig. 7. Measured TCP paths displayed in RobotStudio.

The four individual cycle times add up to the total recorded by the batch clock. Table XV summarizes the batch performance. The reported collision result applies only to the objects and exclusions specified in Section VI-C.

B. Effect of Packing Position

The cycle times for slots 1 and 3 are similar, while slots 2 and 4 take approximately 1.0 s longer. The latter two slots are located at $y = 620$ mm rather than $y = 350$ mm. However, because this batch contains only one observation per slot, differences in the incoming pipe poses may also affect the measured times.

C. Placement Error

The fixture and box release errors are summarized in Table XVI. For part 3, the final height in the box is approximately 300.587 mm, compared with a target height of 300 mm. The recorded positional error for this release is 0.587 mm.

Because the simulation assumes exact vision poses and rigid attachment, these errors mainly reflect the numerical placement results at the sampled release instants. They should not be interpreted as measurements of real camera accuracy or robot hardware accuracy.

TABLE XV
BATCH SUMMARY

Metric	Result
Completed parts / boxes	4 / 1
Total batch time	150.596 s
Mean cycle time	37.649 s
Minimum / maximum cycle	37.088 / 38.252 s
Calculated throughput	95.62 parts/h
Nominal targets with solutions	21 / 21
RAPID program-check errors	0
Fixture / box release events	4 / 4
Simulation steps	6,286
Collision samples within defined checks	0

TABLE XVI
POSITION ERROR STATISTICS

Metric	Fixture (mm)	Box (mm)
Mean error	0.183596	0.204399
RMS error	0.299351	0.309239
Maximum error	0.577977	0.586858

D. Time Breakdown and Improvement

Table XVII breaks down the explicit waits included in the timed batch. Each part goes through nine state handshakes: states 5, 8, 10, 20, 25, 30, 40, 45, and 50.

Together, these waits account for approximately 15.7% of the total batch time. The remaining time is spent on robot motion, fine positioning, and program execution, suggesting that the transfer paths offer more room for improvement than the fixed waits.

Our current motion plan uses fixed clearance offsets. Possible improvements include lowering selected clearance targets, shortening travel distances, and adjusting transition zones. Any revised plan should be evaluated with the same input seed to allow a fair comparison. We have not yet measured an optimized version, so no improvement in cycle time is claimed here.

VIII. CONCLUSION

The final RAPID module was loaded into the virtual controller and passed the program check without errors. We also unpacked the final package into a separate folder and successfully restored its controller backup. In the completed simulation, the conveyor feeds and stops each pipe for pickup, the robot inserts the pipe horizontally into the wall port, and a second grasp allows it to withdraw the pipe. The vision input is idealized, suction attachment is modeled as rigid, and

TABLE XVII
EXPLICIT WAITS INSIDE THE TIMED BATCH

Wait group	Calculation	Total (s)
Gasket process	4×2	8.0
Belt feed	4×3	12.0
State handshakes	$4 \times 9 \times 0.10$	3.6
All listed waits		23.6
Other timed activity	$150.596 - 23.6$	126.996

the gasket operation is represented by a timed wait. These simplifications allow us to focus on the handling sequence required by the assignment.

Overall, we built and tested a two-grasp packing cell using calculated surface contact poses and indexed box placement. The final batch processed four parts in 150.596 s, corresponding to a calculated throughput of 95.62 parts/h. The results demonstrate that the modeled geometry and control sequence operate together as intended in simulation. The packing paths to the farther row took longer in this batch, indicating that clearance heights and travel distances are worth reviewing before attempting to increase the robot speed.

IX. LIST AND PHOTO OF MEMBERS

Group Name: Projabez

Group Members: Guo Rongen (left in the first photo), Yang Jiaxin (centre of the first photo), Han Peitong (right in the first photo), and Hong Zhuoyi (in the second photo).



Fig. 8. Group Image 1



Fig. 9. Group Image 2